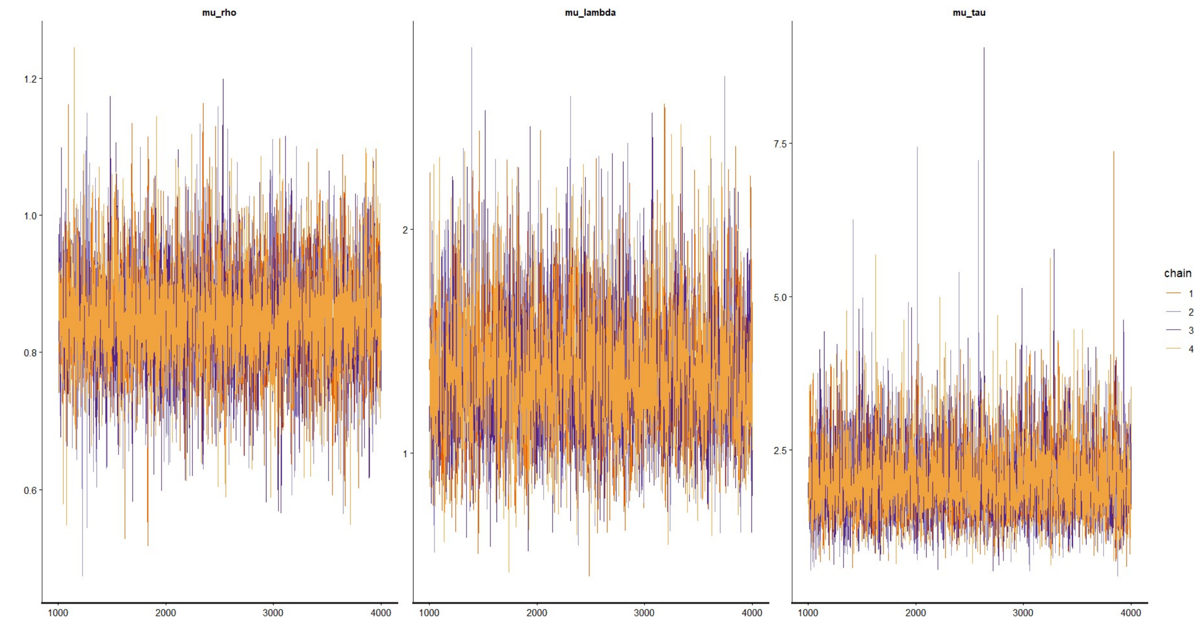
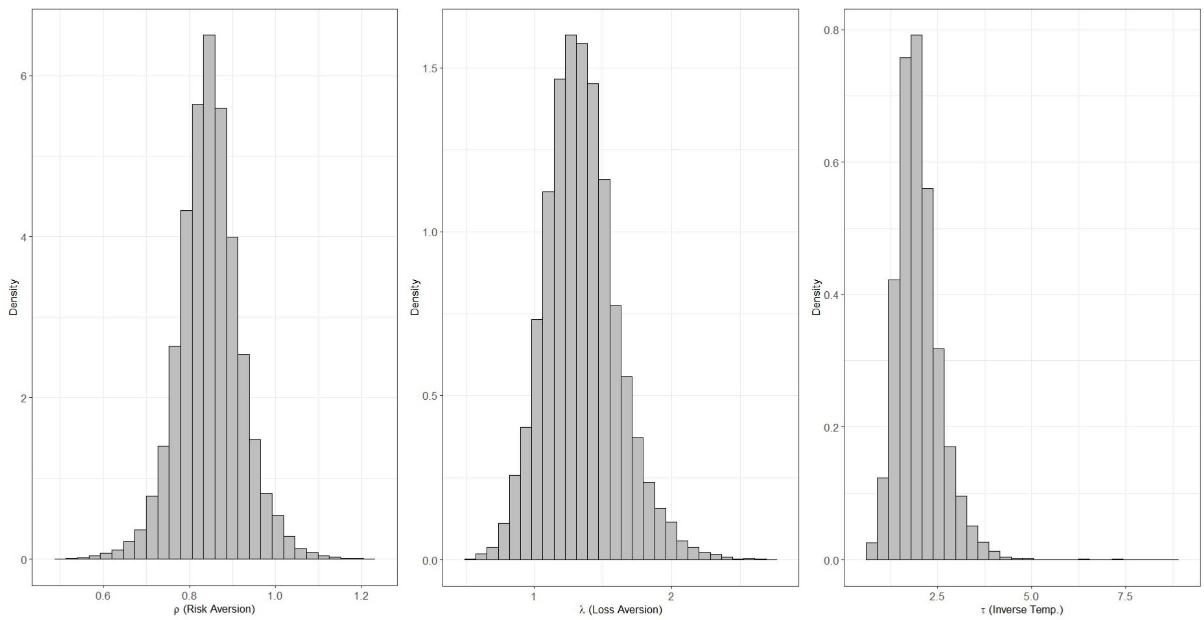
Q1-(1). I used the hBayesDM package and done MCMC with ra_prospect model. Convergence of the four chains is presented below.

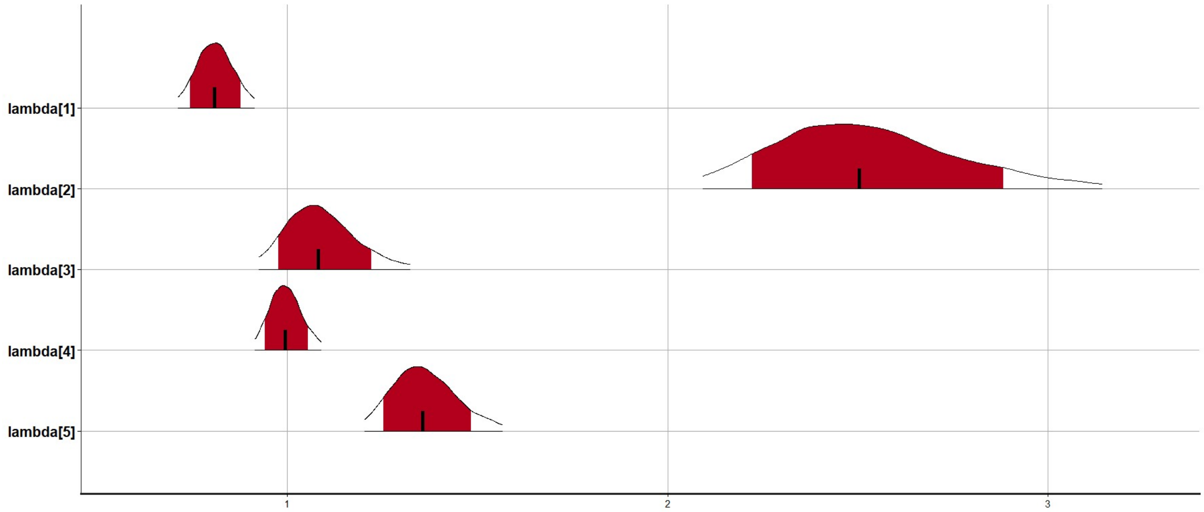


Rhat values of each chains did not exceed 1.1, settled near 1. Minimum value was 0.9997562 and the maximum was 1.0008709.

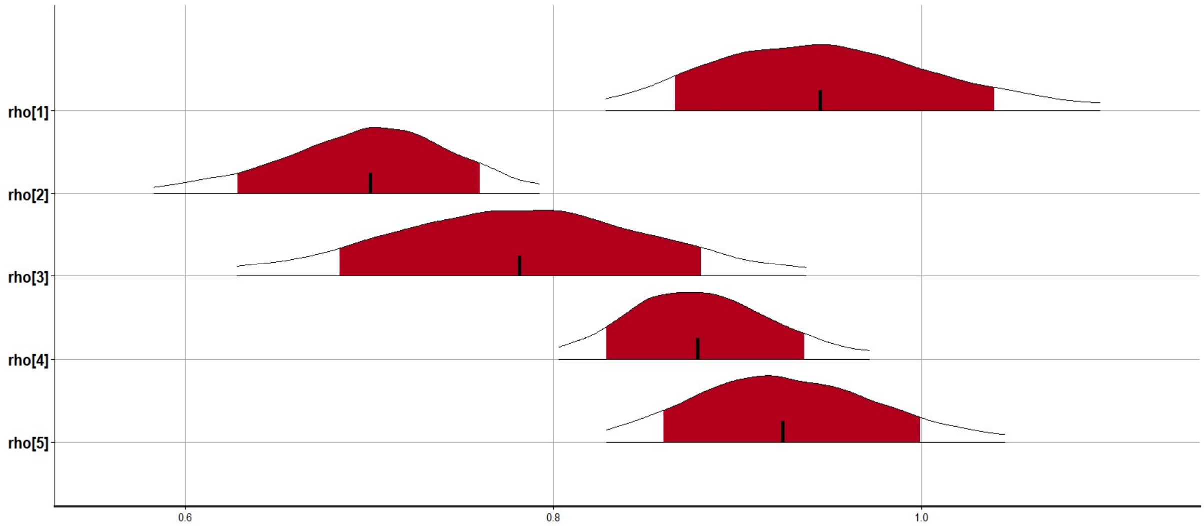
Even though it shows some peaks in parameter tau, they fairly look like 'hairy caterpillars'. Estimated posterior distributions are presented as below.



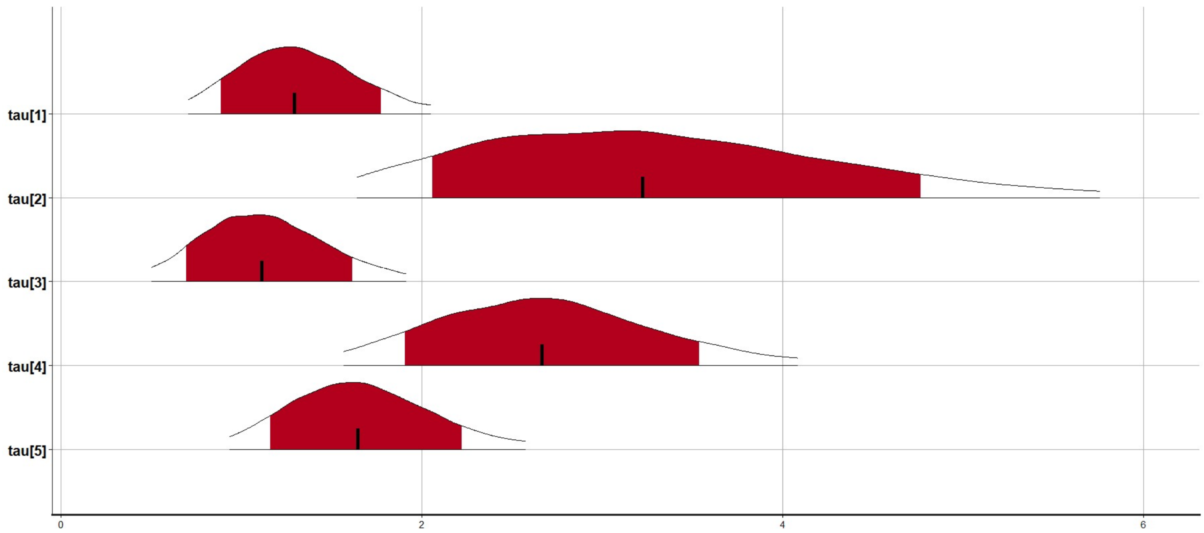
This is a posterior distribution of hyper-parameter of the prospect model. For each subject, the posterior distribution looks like this.



These are posterior distributions of lambda of each subjects.



These are posterior distributions of each rho of each participants.

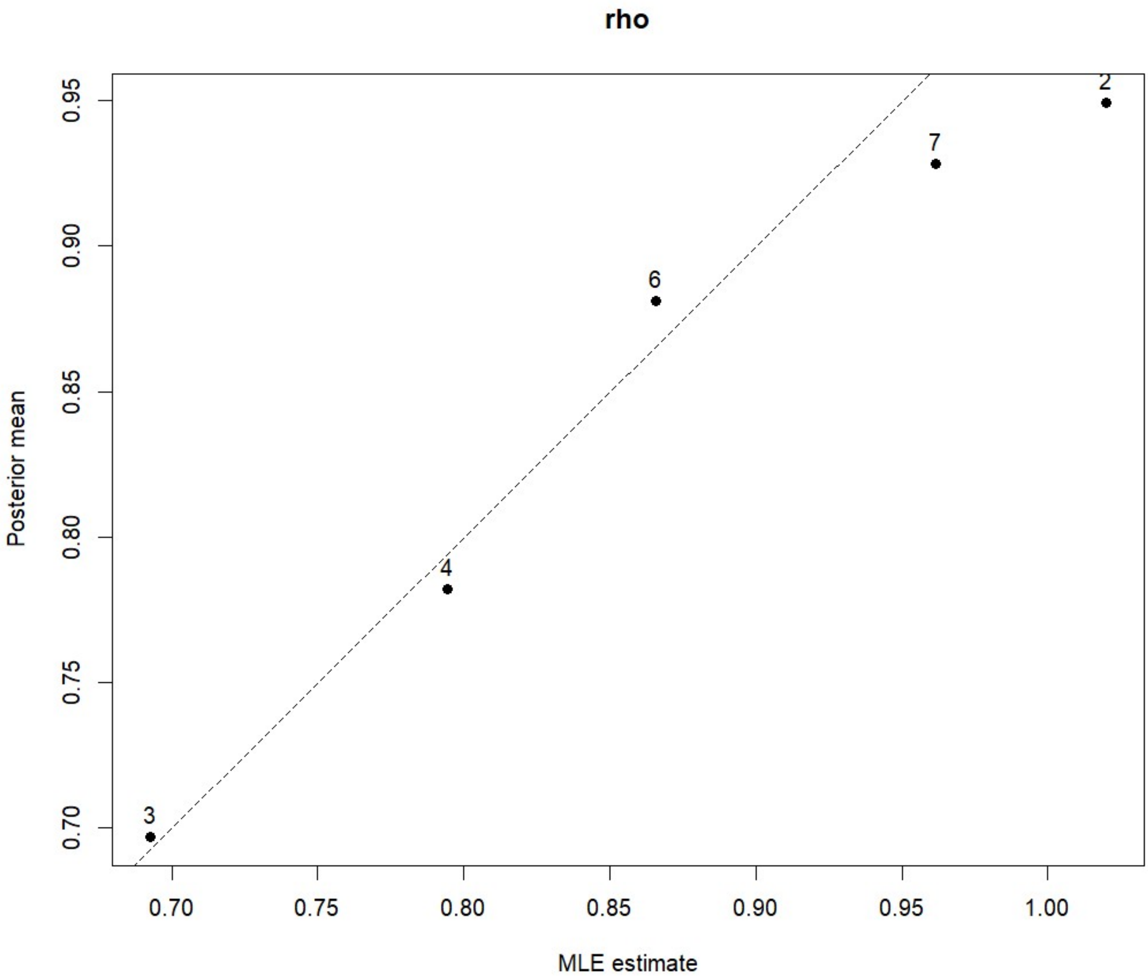


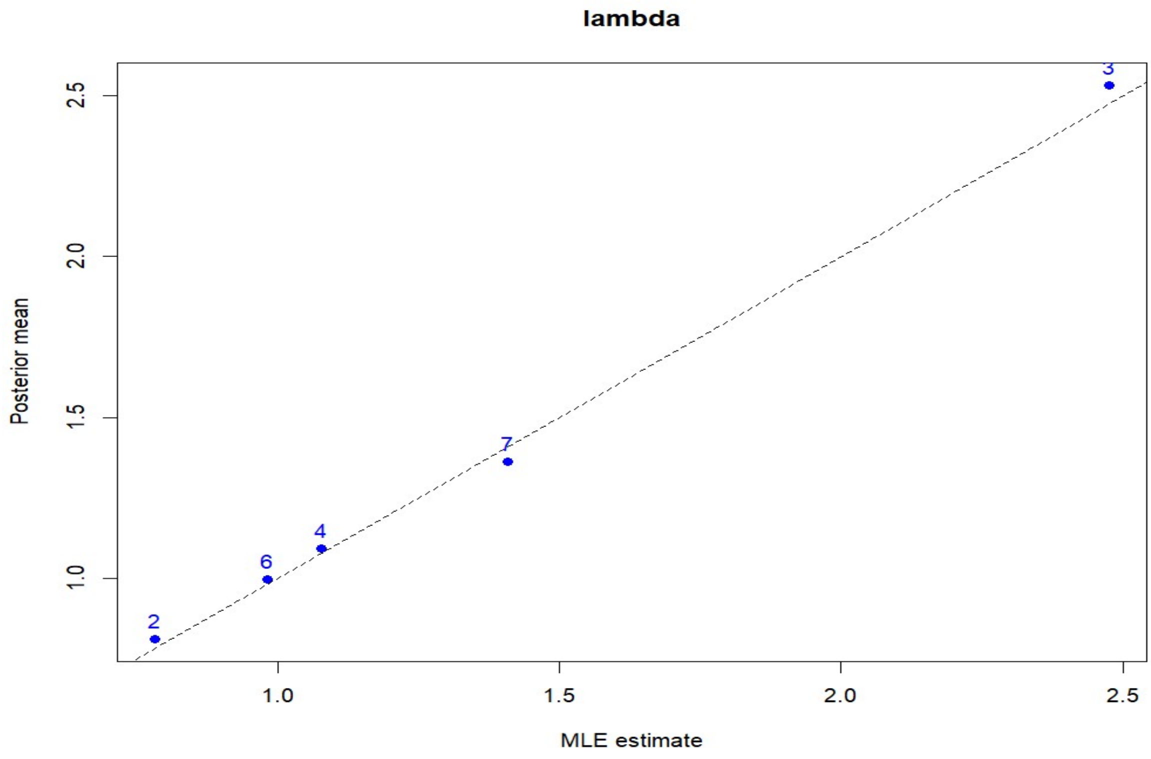
These are posterior distributions of tau of each subjects.

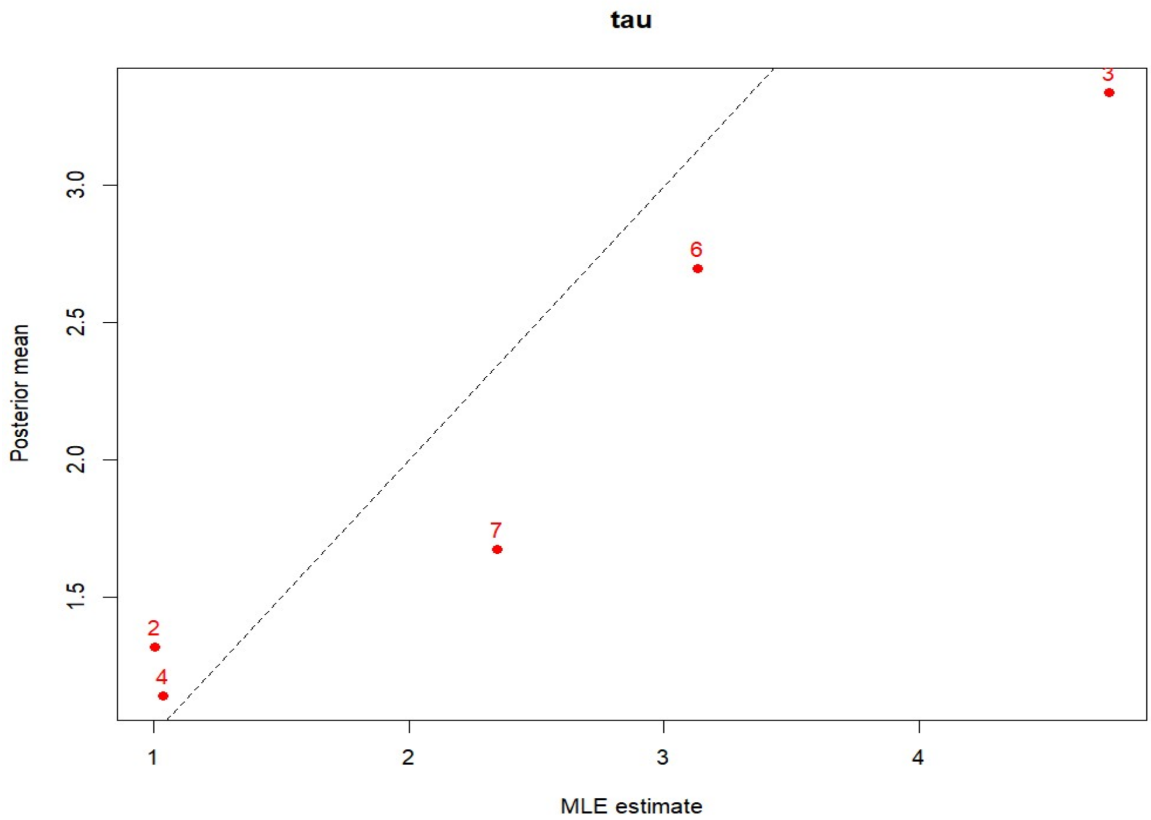
Comparing MLE and posterior mean is done in a table below.

Subject ID	MLE estimates			Posterior mean		
	rho	lambda	tau	rho	lambda	tau
2	1.0199	0.7828	1.0071	0.949470	0.81001382	1.314553
3	0.6927	2.4747	4.7421	0.6961486	2.5363474	3.326648
4	0.7944	1.0765	1.0355	0.7821422	1.0930051	1.134298
6	0.8656	0.9823	3.1299	0.8803819	0.9936756	2.698912
7	0.9616	1.4085	2.3468	0.9289882	1.3623532	1.670813

Q1-(2). Visually comparing individual parameter values are presented below.







The dashed lines are x=y lines. Whie rho and lambda show quite similar values, tau is seemingly quite different.

Q1-(3). MLE estimates the parameter values that are most likely generated the data. However, the hBayesDM uses the MCMC to estimate posterior distributions of each parameters. And the plotted dots of y-axis are posterior means. In addition, hierarchical Bayesian model induces shrinkage, which leads individual level parameters pulled toward the group distribution. These are the reasons of different results in two approaches.

Q2-(a). In HW#2, BIC values of ra_prospect (model 1) and ra_noLA (model 2) were 580.4341 and 651.8656, respectively. Since $BF_{12} = \exp\left(-0.5[(BIC_1) - (BIC_2)]\right)$ as presented in the textbook, $BF_{12} = \exp\left(-0.5[(580.4341) - (651.8656)]\right)$. $BF_{12} = \exp(35.71575) ≒ 3.24454 \times 10^{15}$.

Q2-(b). This value is much larger than 100, and it means extreme evidence for model 1, according to Jeffery' evidence categories.

Q2-(c). Posterior odds can be calculated by multiplying prior odds and Bayes factor. Prior odds are p(Model 1)/p(Model 2), and in this case 1.5. Posterior odds are $BF_{12} \times 1.5$ ≒ 4.86681×10^{15}.

Q2-(d). p(M1|D) = 1 – p(M2|D), and posterior odds = p(M1|D)/p(M2|D) ≒ 4.86681 $\times 10^{15}$. p(M1|D) = posterior odds × (1-p(M1|D)) = posterior odds – (posterior odds)(p(M1|D). p(M1|D)(posterior odds + 1) = posterior odds. P(M1|D) = posterior odds/(1 + posterior odds). So, p(M1|D), which is posterior probability of model 1, can be calculated $\frac{4.86681 \times 10^{15}}{4.86681 \times 10^{15} + 1}$, which is 0.99999999999999979452659955905412, nearly 1.

Q2-(e). The BIC values of memory retention models are presented in the table below.

Models	BIC
POW1 (model 1)	475.292
POW2 (model 2)	477.371
EXP1 (model 3)	524.532
EXP2 (model 4)	489.335
EXPOW (model 5)	479.451
HYP1 (model 6)	482.954
HYP2 (model 7)	481.634

To compare these models, I will compare POW1 with other models, since it has the smallest BIC value.

$BF_{12} = \exp\left(-0.5[(475.292) - (477.371)]\right)$

$BF_{13} = \exp\left(-0.5[(475.292) - (524.532)]\right)$

$BF_{14} = \exp\left(-0.5[(475.292) - (489.335)]\right)$

$BF_{15} = \exp\left(-0.5[(475.292) - (479.451)]\right)$

$BF_{16} = \exp\left(-0.5[(475.292) - (482.954)]\right)$

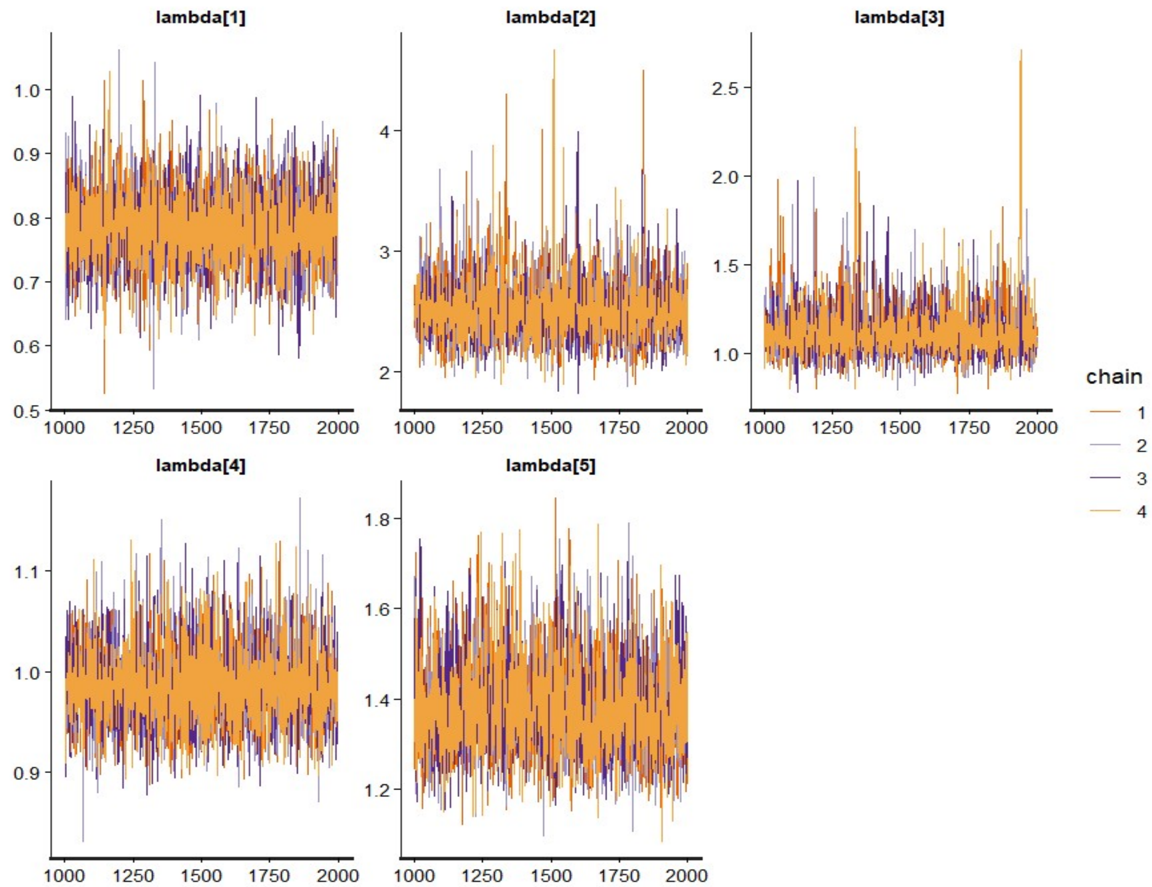
$BF_{17} = \exp\left(-0.5[(475.292) - (481.634)]\right)$

Calculated values are presented below.

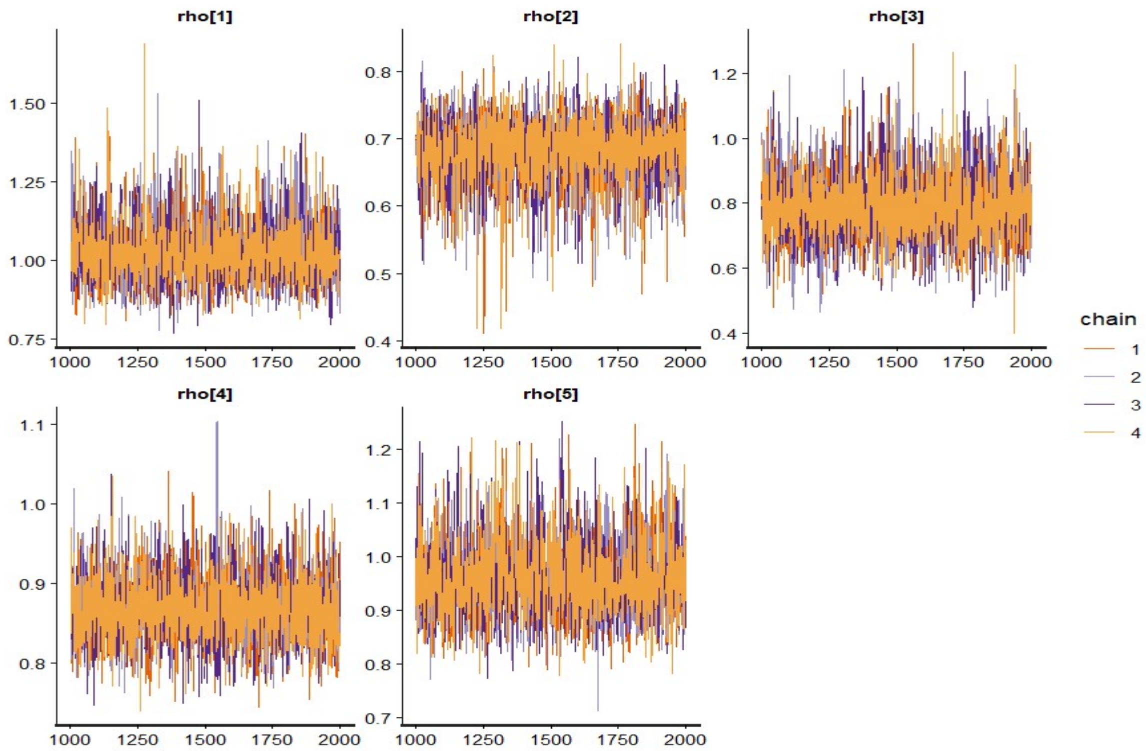
BF_{12}	BF_{13}	BF_{14}	BF_{15}	BF_{16}	BF_{17}
2.827	4.924×10^{10}	1120.466	7.999	46.109	23.831

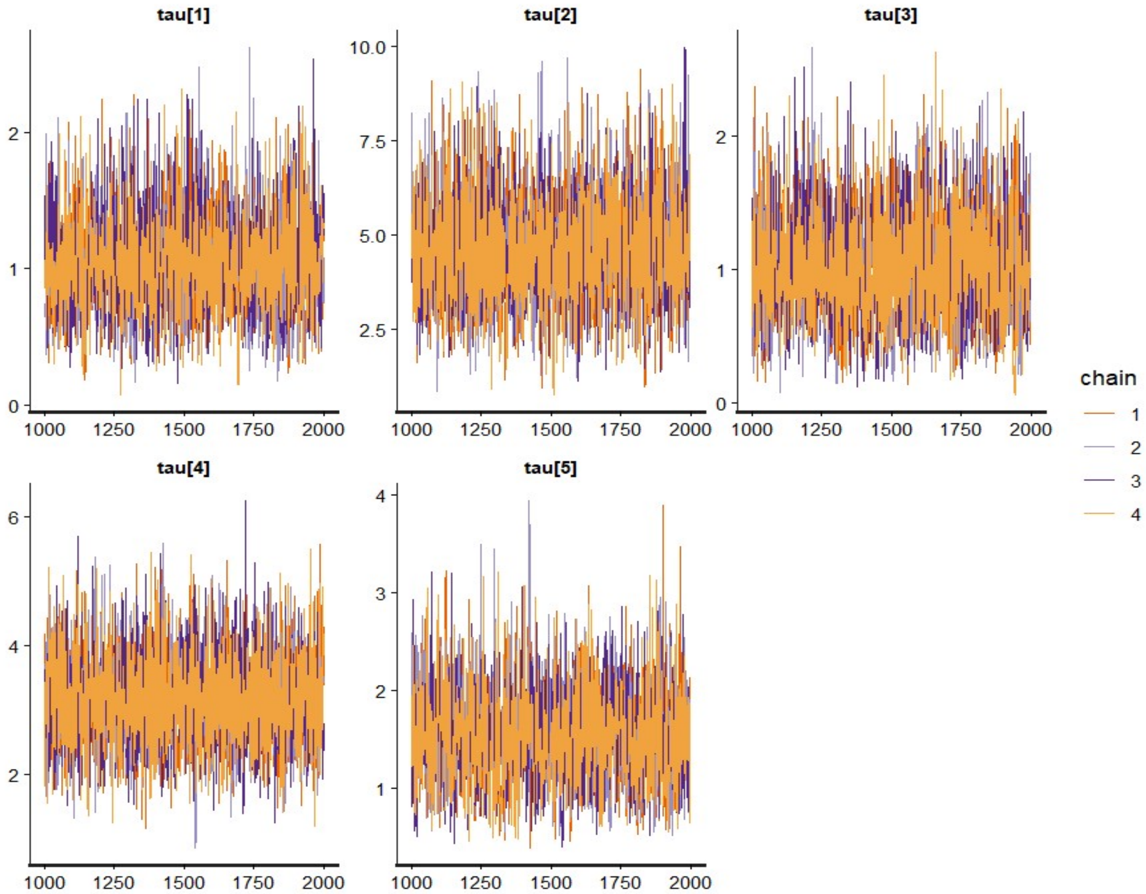
This result means that model 1, POW1 model is more supported than other models.

Q3-(1). I modified the code into ra_prospect_stan_allSubj.stan and conducted MCMC with it. The pictures below shows the trace plot.

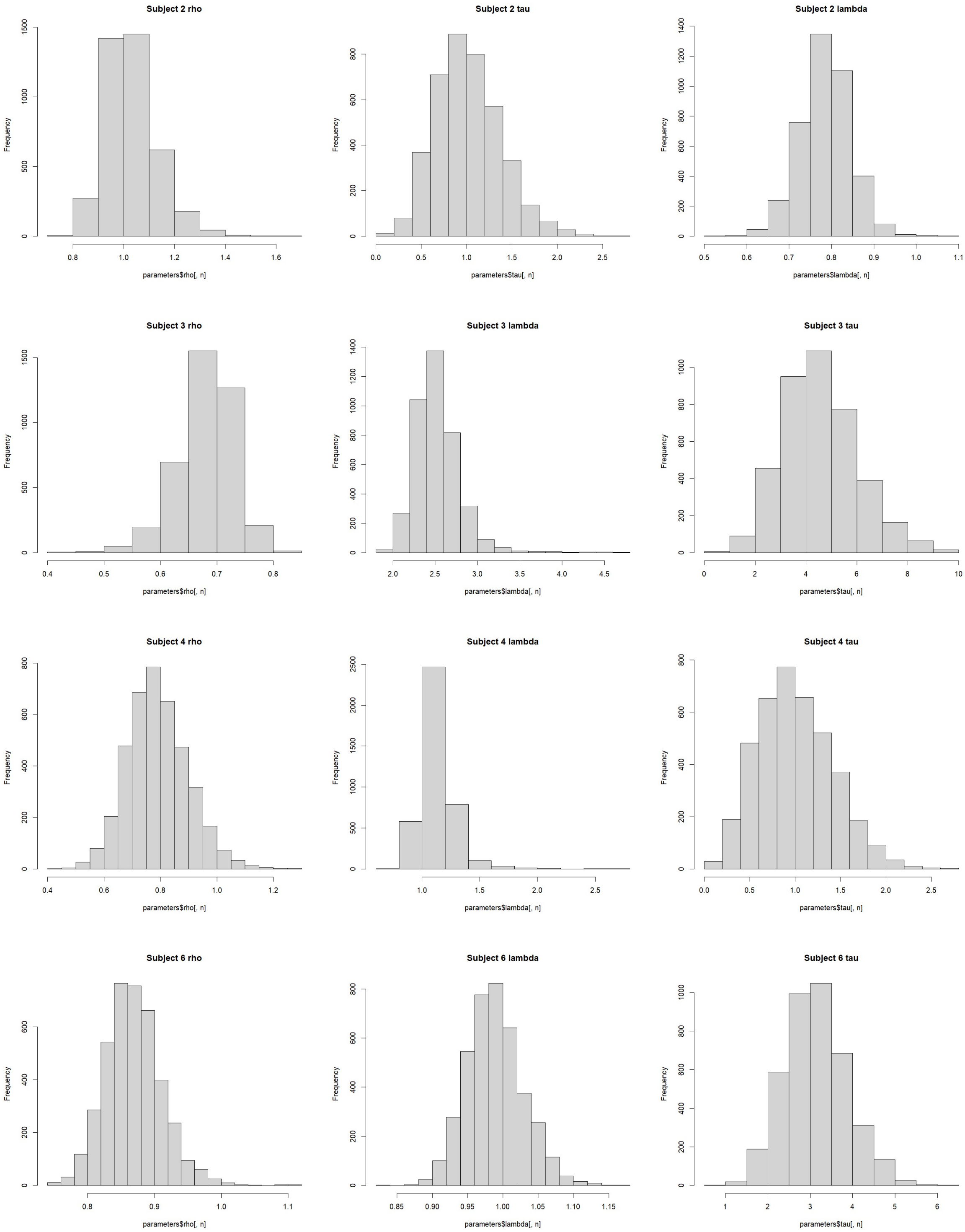


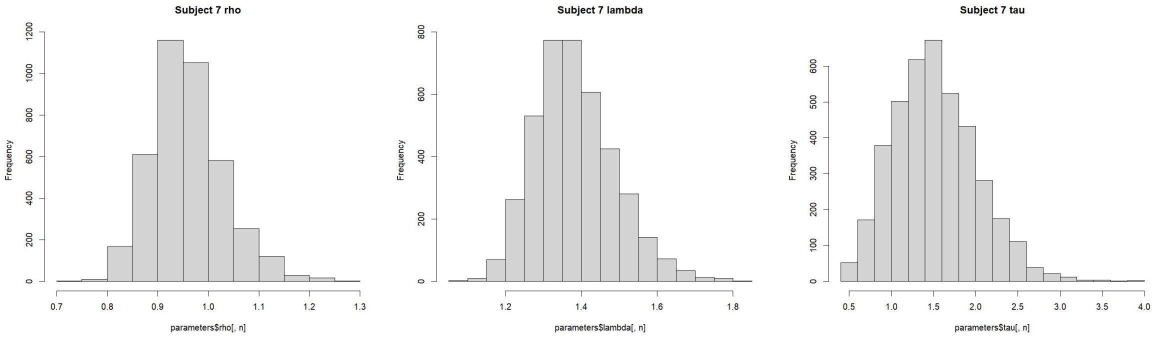
Even though they show spikes in some regions, the convergence seems to be acceptable. Rhat values were all equally 1, which shows a good convergence.


rho[1]
rho[2]
rho[3]
rho[4]
rho[5]
chain
1
2
3
4
1000 1250 1500 1750 2000


tau[1]
tau[2]
tau[3]
tau[4]
tau[5]
chain
1
2
3
4
1000 1250 1500 1750 2000

Plotted posterior distributions of all subjects are presented in the pictures below.





Q3-(2). I calculated posterior means of all subjects and parameters with using colMeans(). The table below shows the results.

Subject ID	rho	lambda	tau
2	1.0297	0.7856	1.0161
3	0.6806	2.5245	4.5546
4	0.7903	1.1307	1.0012
6	0.8687	0.9888	3.1268
7	0.9588	1.3831	1.5201